

CEP SRS DOCUMENT

Course: Software Design and Architecture

“Career Connect”

(Job Portal)

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ABSTRACT

The "Job Portal" project, named Career Connect, aims to create a platform where job seekers and employers can easily connect. This online system provides job seekers with tools to find suitable jobs and apply for them, while employers can post job openings and manage applications in one place. The platform uses the ASP.NET MVC framework to ensure reliability, security, and a user-friendly experience.

This project solves the inefficiencies in existing job portals by improving the overall usability, offering a responsive interface, and adding features like advanced search options and application tracking. By streamlining the job search and hiring process, this system provides value to both job seekers and employers.

TABLE OF CONTENTS

ABSTRACT

1.	INTRODUCTION TO PROBLEM	9
1.1	Introduction	9
	PROBLEM STATEMENT	10
1.2	Purpose	10
	PROJECT SIGNIFICANCE	10
1.3	Objective	11
1.4	Existing solution	11
1.5	Purposed Solution	11
1.6	Scope	12
2	LITERATURE REVIEW	13
3	PROBLEM DEFINITION & REQUIREMENTS ANALYSIS	14
3.1	Problem Definition	14
3.2	Requirement Analysis	15
4	DESIGN & IMPLEMENTATION	16
4.1	Design	17
4.2	Implementation	17
5	TESTING	18
5.1	Testing Procedure & Test Cases:	19
5.2	Results	19
6	FUTURE ENHANCEMENTS	21
6.1	Future Enhancements	21

Appendix A: Bibliography / References

Appendix B: Index

CHAPTER # 1
INTRODUCTION TO PROBLEM

1. INTRODUCTION TO PROBLEM

1.1 Introduction

The "Job Portal" system, Career Connect, is designed to make job searching and hiring more efficient. It provides an online space where job seekers can look for positions that match their skills and preferences, and employers can post job vacancies and manage applications. By using the ASP.NET MVC framework, this project ensures a reliable and smooth user experience.

Problem Statement

Existing job portals in Pakistan often fall short when it comes to usability, responsiveness, and effective application management. This leads to frustration for users, making it harder for job seekers to find relevant opportunities and for employers to attract suitable candidates.

1.2 Purpose

The main goal of this project is to build a comprehensive platform that addresses the challenges of existing job portals. By providing a simple, efficient, and responsive system, Career Connect will enhance the experience for all users.

Project Significance

Career Connect will be developed using ASP.NET MVC, with a focus on creating an intuitive interface that works seamlessly on all devices. The platform aims to solve usability issues and improve the process of job searching and hiring..

1.3 Objective

- To create a space where employers can post job openings and manage applications.
- To enable job seekers to search for and apply to relevant positions easily.
- To simplify and streamline the hiring process.

1.4 Existing solution

While there are several job portals available, they often lack a user-friendly design, take too long to load, and do not provide efficient tools for managing job applications. As a result, users may leave these platforms without achieving their goals.

1.5 Purposed Solution

Career Connect will address these issues by offering a modern, responsive design, advanced job search features, user profile management, and a clear application tracking system. The platform will provide a seamless experience for both employers and job seekers.

1.6 Scope

The system will allow job seekers to find opportunities and employers to manage postings efficiently. The platform will be accessible on desktops, tablets, and mobile devices. Future enhancements could include AI-driven job recommendations and analytics for tracking hiring trends.

CHAPTER # 2
Literature Review

2 Literature Review

This chapter examines existing job portals, their features, and limitations. It also explores how modern technologies, such as ASP.NET MVC, can be used to overcome these limitations.

Background Study

Job portals have become essential tools for connecting employers and job seekers. However, many existing systems fail to deliver a satisfactory user experience. Common issues include slow load times, non-responsive designs, and limited search functionalities.

Issues with Current Portals

1. **User Interface Problems:** Many job portals lack intuitive designs, making them difficult to navigate. Users often struggle to find relevant information quickly.
2. **Limited Functionality:** Features like advanced search filters, personalized recommendations, and real-time notifications are often missing, reducing the overall effectiveness of these platforms.
3. **Technical Challenges:** Slow page loading, lack of mobile responsiveness, and outdated frameworks lead to a subpar experience.

Advances in Job Portal Development

Modern web development frameworks like ASP.NET MVC offer solutions to these challenges by enabling faster, more secure, and scalable applications. These technologies allow for:

- Enhanced user experiences with responsive designs.
- Advanced features such as customizable dashboards and real-time data updates.
- Efficient data management and application tracking systems.

By learning from the shortcomings of existing systems, Career Connect aims to provide a robust and user-friendly alternative that meets the needs of job seekers and employers effectively.

CHAPTER # 3

Problem definition and requirement analysis

3 Problem Definition & requirements analysis

3.1 Problem Definition

Career Connect addresses the inefficiencies in existing job portals by creating a feature- rich, responsive platform. The system will make it easier for job seekers to find suitable jobs and for employers to hire qualified candidates.

3.2 Requirement Analysis

Functional Requirements:

- User roles: Admin, Job Seekers, and Employers.
- Job posting and application tracking features.
- Advanced search and filtering options.
- Profile creation and management for all users.

Non-functional Requirements:

- Data security and integrity.
- Mobile-friendly, responsive design.
- Scalability to support a growing number of users.

CHAPTER # 4

Design & Implementation

4 DESIGN & IMPLEMENTATION

4.1 Design

Career Connect is built using the ASP.NET MVC framework. The design includes:

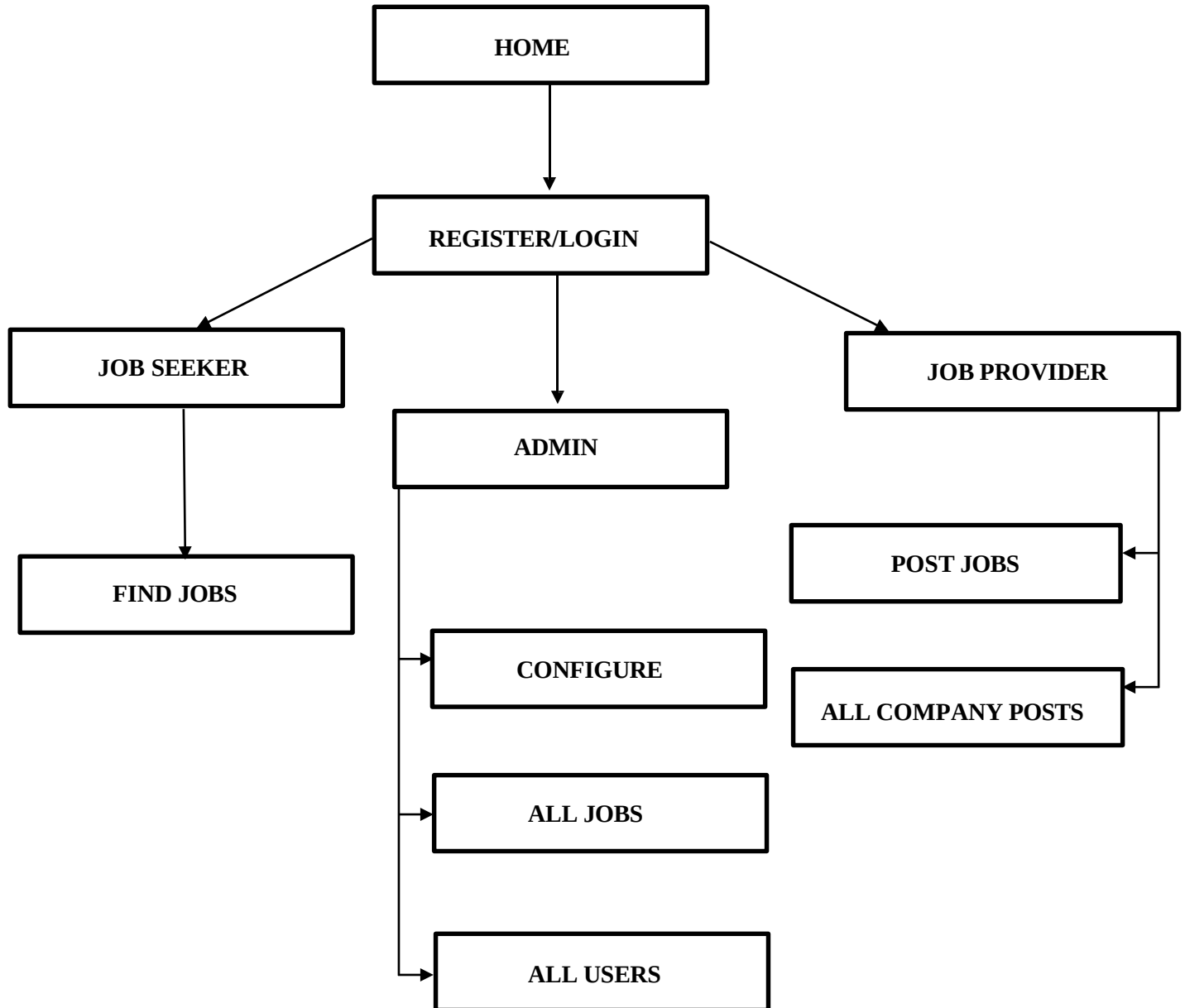
- Database: Tables for storing user details, job postings, and applications.
- User Interface: A clean and responsive design implemented with HTML, CSS, and JavaScript.
- Architecture: Separation of concerns using MVC principles.

4.2 Implementation

The implementation process involves:

- Setting up a database schema for storing data securely.
- Creating Razor views for a dynamic front-end.
- Implementing controllers to handle user interactions.
- Testing and refining the system before deployment.

4.3 Data Flow Chart



CHAPTER # 5

Tests & Results

5 TESTING

5.1 Testing Procedure & Test Cases

Testing methods include:

- Unit Testing: Testing individual components to ensure they work correctly.
- Integration Testing: Ensuring all components work together seamlessly.
- User Testing: Collecting feedback to validate the system's usability and performance.

5.2 Results

The system meets the functional and non-functional requirements. All tests confirm that the platform operates as intended, with no major issues.

CHAPTER # 6

Future Enhancements

6 Future Enhancements

- **AI Features:** Personalized job recommendations and smart filtering.
- **Mobile Applications:** Native apps for iOS and Android devices.
- **Advanced Analytics:** Detailed insights into hiring trends and user behavior.

CHAPTER # 7

Appendix

Appendix A

- ASP.NET MVC Documentation
- Principles of Software Design and Architecture
- Usability Guidelines for Web Applications

Appendix B

The index provides quick navigation to the report's key sections, making it easy for readers to find relevant information. Below are the categories included in the index:

- **Abstract:** Overview of the project's purpose and scope.
- **Introduction to Problem:** Detailed explanation of the project's background, objectives, and significance.
- **Literature Review:** Examination of current job portal limitations and advances in the field.
- **Problem Definition & Requirements Analysis:** Clarification of system requirements and problem-solving approaches.
- **Design & Implementation:** Insights into the system's architecture, database design, and development process.
- **Testing:** Information on the testing procedures and results.
- **Future Enhancements:** Potential areas for further development and scalability.
- **Bibliography / References:** List of resources and materials referenced during the project.
- **Figures and Diagrams:** Visual aids to support the text, including flowcharts, diagrams, and tables.